Practice workbook · Expressions for file size and transfer time

Building (not evaluating) an expression for the size of an image, sound or text file, how many files fit in a store, or how long a file takes to send at a given bit rate, with all unit conversions.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-11

1. Question 2 · part d.ii · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d.ii only.

2 Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

(2)

(ii) Identify the number of characters that can be represented using standard ASCII.

(1)

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

(2)

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

(2)

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1. Question 2 · part d.ii · 4 marks

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Worked-example retry · Attempt part d.ii only.

(c) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	1	0	0
0	0	0	1	0	1	1	0

(d) A bitmap image is made up of pixels.

(i) An image has five colours.

Complete the table by adding a unique binary pattern for each colour.

Each pattern must use the **same minimum colour depth**.

(2)

Colour	Binary pattern
Green	
Black	
White	
Red	
Blue	

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1. Question 2 · part d.ii · 4 marks

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Worked-example retry · Attempt part d.ii only.

(ii) Another image is 3579 pixels high and 6128 pixels wide.

The image is stored with a 32-bit colour depth.

The metadata for the image is 732 bytes.

Construct an expression to show how the file size, in **megabytes**, is calculated.

You do **not** need to do the calculation.

(4)

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(Total for Question 2 = 15 marks)

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1. Question 2 · part d.ii · 4 marks · Working and answer

Working and answer

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2. Question 4 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

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4 One function of an operating system is file management.

(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

(2)

(b) An image is 2322 pixels high and 4128 pixels wide.

The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

(4)

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2. Question 4 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(c) Operating systems often include compression software for reducing file sizes.

(i) Give **two** reasons for reducing file sizes.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** drawbacks of using compression software.

(2)

1 .....

.....

2 .....

.....

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12

P 6 1 8 8 0 A 0 1 2 2 4

2. Question 4 · part b · 4 marks · Working and answer

Working and answer

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3. Question 4 · part d · 4 marks

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Further practice · Attempt part d only.

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(a) Both kibibyte and kilobyte can be used as measures of file size.

Compare kibibyte and kilobyte.

(2)

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The image is stored with a 16-bit colour depth.

The metadata for the image is 975 bytes.

Construct an expression to show how the file size, in megabytes, is calculated.

You do **not** need to do the calculation.

(4)

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3. Question 4 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

(c) Operating systems often include compression software for reducing file sizes.

(i) Give **two** reasons for reducing file sizes.

(2)

1 .....

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2 .....

.....

(ii) Give **two** drawbacks of using compression software.

(2)

1 .....

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2 .....

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3. Question 4 · part d · 4 marks

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Further practice · Attempt part d only.

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(d) A text file is stored on a hard disc.

The file holds information from one side of a sheet of paper.

The sheet of paper is represented as a grid, 80 columns wide and 66 rows long.

Each cell in the grid contains a single 2-byte Unicode character.

The file also contains 40 characters of metadata.

The hard disc allocates space in blocks of 1024 bytes.

Construct an expression to show the number of blocks required to store the file.

You do **not** need to do the calculation.

(4)

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(Total for Question 4 = 14 marks)

3. Question 4 · part d · 4 marks · Working and answer

Working and answer

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4. Question 4 · part b · 4 marks

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Further practice · Attempt part b only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

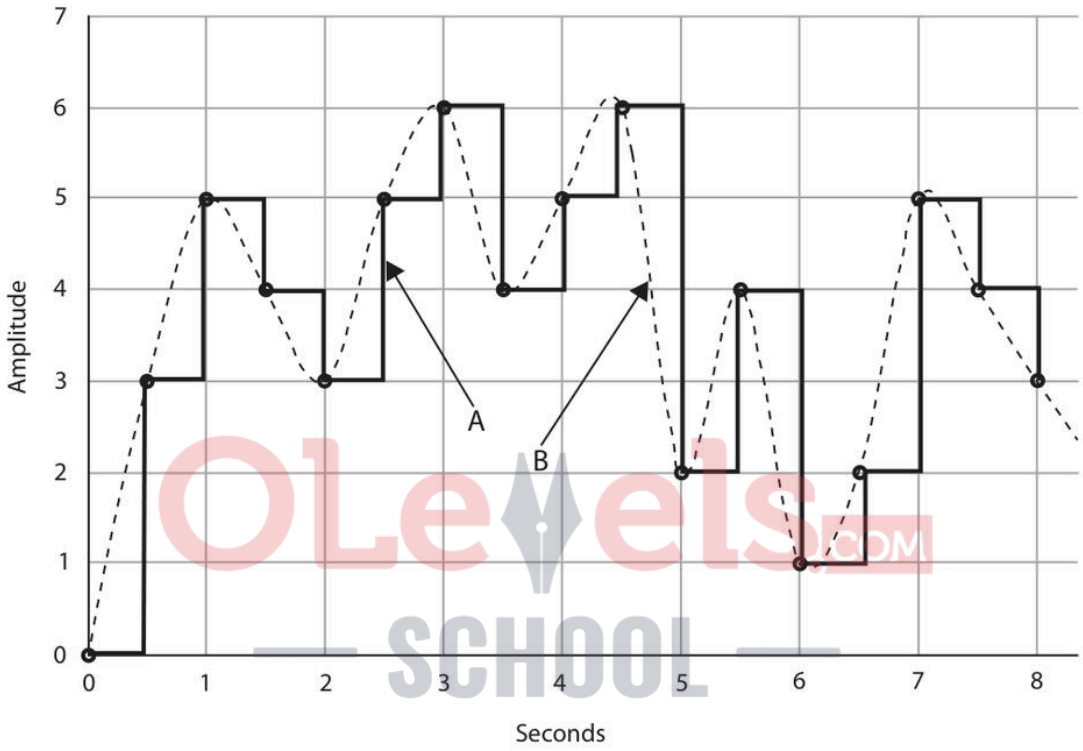


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

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P 6 1 8 8 4 R R A 0 1 0 2 0

4. Question 4 · part b · 4 marks

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Further practice · Attempt part b only.

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(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

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(vi) Give the value in binary of the audio sample at the 4th second.

(1)

4. Question 4 · part b · 4 marks

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Further practice · Attempt part b only.

(b) Ibrahim wants to store a music file.

- The sampling frequency is 47 kHz.
- The bit-depth is 64
- The analogue signal lasts for 819 seconds.

Construct an expression to show how the file size, in bytes, is calculated.

You do **not** need to do the calculation.

(4)

(c) Ibrahim wants to upload a music file to his online storage. The file has been compressed using a lossy algorithm.

Give **one** drawback of using a lossy algorithm for this purpose.

(1)

(Total for Question 4 = 12 marks)

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4. Question 4 · part b · 4 marks · Working and answer

Working and answer

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5. Question 1 · part g.ii · 1 mark

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Further practice · Attempt part g.ii only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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5. Question 1 · part g.ii · 1 mark

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Further practice · Attempt part g.ii only.

(d) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	0	1	0
0	0	1	1	0	0	1	1

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

(1)

(f) Give the number of bits per character used by standard ASCII.

(1)

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

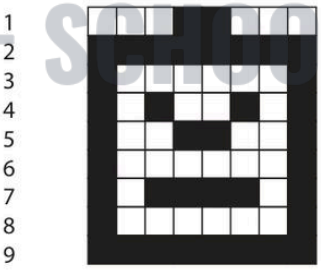


Figure 1

(i) State what is meant by the term **pixel**.

(1)

(ii) Construct an expression to show the number of pixels in the image.

(1)

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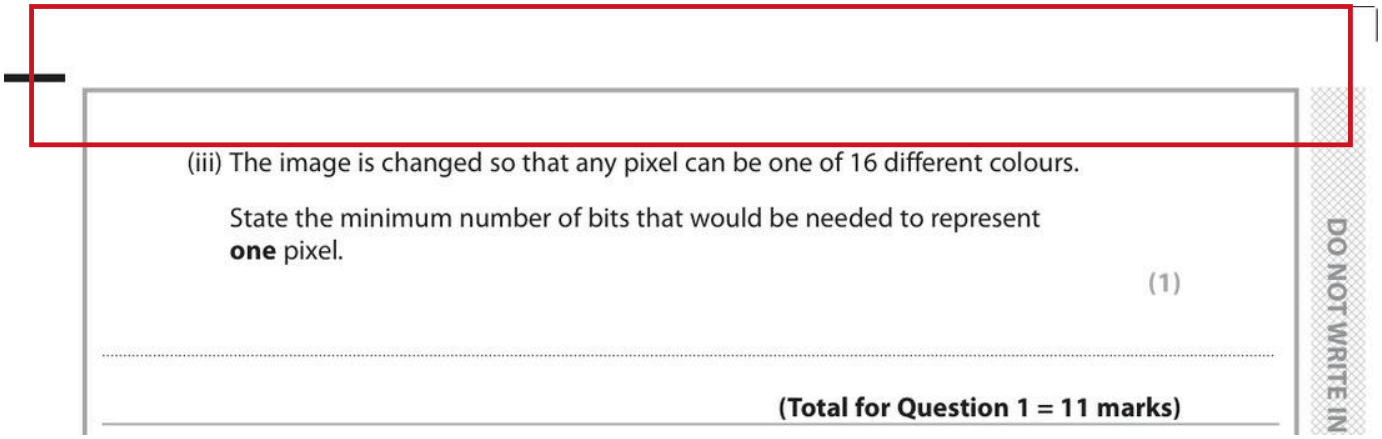
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5. Question 1 · part g.ii · 1 mark

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Further practice · Attempt part g.ii only.



(iii) The image is changed so that any pixel can be one of 16 different colours.

State the minimum number of bits that would be needed to represent **one** pixel.

(1)

.....

(Total for Question 1 = 11 marks)

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Working and answer

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6. Question 2 · part g · 4 marks

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Further practice · Attempt part g only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

6. Question 2 · part g · 4 marks

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Further practice · Attempt part g only.

(e) **Figure 2** shows a network topology.

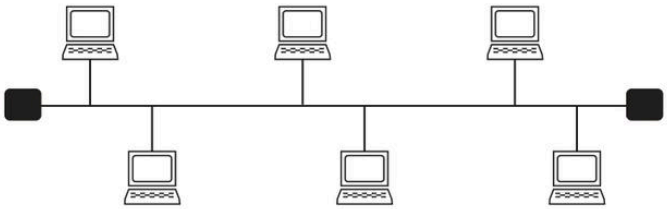


Figure 2

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1

2

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6

P 6 6 4 8 9 A 0 6 2 0

6. Question 2 · part g · 4 marks

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Further practice · Attempt part g only.

(ii) Identify which **one** of these file formats uses lossy compression.

(1)

- ☒ **A** BMP
- ☒ **B** DOC
- ☒ **C** JPEG
- ☒ **D** PNG

(iii) Here is a string of data.

CCCWWWCCWWWWWWCCC

Give the result of compressing the string using a run-length encoding algorithm.

(1)

.....

.....

(g) A video file is to be transmitted over the internet.

- The network transmission speed is 54 Mbps.
- The file size is 6 gigabytes (GB).

Construct an expression to show how the transmission time, in seconds, is calculated.

You do **not** have to do the calculation.

(4)

(Total for Question 2 = 18 marks)

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6. Question 2 · part g · 4 marks · Working and answer

Working and answer

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7. Question 1 · part b.i · 2 marks

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Further practice · Attempt part b.i only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

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(iii) Convert the binary pattern 11000110 to hexadecimal.

(2)

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7. Question 1 · part b.i · 2 marks

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Further practice · Attempt part b.i only.

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(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

(2)

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

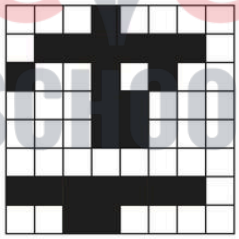


Figure 1

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

(3)

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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7. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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8. Question 3 · part a.ii · 3 marks

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Further practice · Attempt part a.ii only.

3 Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

(1)

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

(3)

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

(1)

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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8. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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9. Question 3 · part b.ii · 4 marks

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Further practice · Attempt part b.ii only.

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3 Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

(2)

(ii) Identify the statement that is true for hexadecimal.

(1)

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

(2)

9. Question 3 · part b.ii · 4 marks

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Further practice · Attempt part b.ii only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

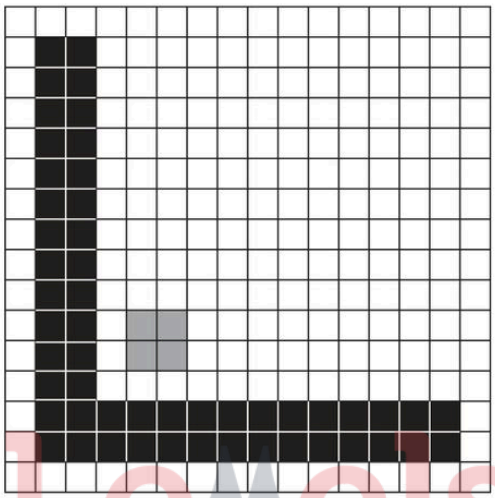


Figure 1

(i) State the smallest component of a bitmap image.

(1)

.....

.....

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

(4)

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9. Question 3 · part b.ii · 4 marks

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Further practice · Attempt part b.ii only.

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(iii) Explain why a run-length encoding algorithm is a good choice for compressing the image in **Figure 1**.

(2)

(c) The **Pigpen** cipher shown in **Figure 2** has been used to encrypt a message.

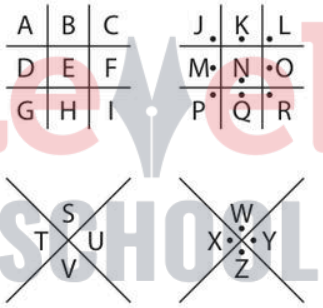
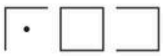


Figure 2

Complete the table to give the decrypted plain text.

(2)

Encrypted text	Plain text
	
	

(Total for Question 3 = 14 marks)

9. Question 3 · part b.ii · 4 marks · Working and answer

Working and answer

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